

Group: 15

Board Member: Dr. Pilar Garcia Souto

Project Title: Development and testing of a data collection system for thermoregulation and thermal comfort studies

Word Count: 3000

Identifier	Roles, Tasks, etc.
{AA}	Communications Officer and Hardware Co-ordinator. Responsible for contacting the board member and course coordinators. Researched the hardware necessary to design and develop the new system. Created the new system and aided in data collection.
{BB}	Participated in laboratory sessions for the prototype creation and the data collection sessions. Created the Python code that formed live graphs of the wired systems. Authored the Results and Analysis section.
{CC}	Acted as Secretary and Literature Co-ordinator. Responsible for formal minutes, report compilation, and LaTeX formatting. Authored the Introduction and coordinated the literature review.
{DD}	Contributed in experimental design of the project, participated in lab sessions and data collection. Wrote up the abstract.
{EE}	Managed financial planning and coordinated gathering of necessary hardware components. Contributed to early laboratory sessions for initial construction of new prototype.

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Chapter 1

Abstract {DD, EE}

Thermal comfort in university spaces, such as libraries and lecture halls, plays a major role in student productivity and well-being. However, stationary thermostats in these facilities and reliance on average population models fail to account for physiological indicators and the micro-climates within different rooms.

This project details the design, prototyping, and evaluation of 'System 2.0', a wireless system intended to provide individualised thermal comfort sensing. The device features high-precision sensors to monitor humidity, surface temperature, ambient temperature, skin temperature and CO_2 concentration.

To validate the system, it was compared against a high-accuracy wired system. Environmental variables were measured across 4 UCL locations: a laboratory, a lecture theatre, and the Student Centre (L2 and L4). Occupant surveys ($n = 29$) were conducted to correlate the measured values with subjective comfort levels.

Both systems were shown to have similar response times. This confirmed that the wireless system was a reliable data collection alternative. The environmental data highlighted distinct micro-climate variations in the four areas of testing. Denser areas like the Student Centre recorded the highest mean concentrations of CO_2 , reaching 802.9 ppm.

The survey data validated the PMV upper thresholds from literature, but also quantified the 'Discursive Gap' between human perception and actual room conditions. Despite the Student Centre having a recorded mean temperature of 25.43 °C, individual estimates of the temperature had extreme variation, ranging from 20 °C to 37 °C. These findings suggested that human perception was a poor tool in monitoring ambient conditions.

This project demonstrates that the wireless prototype can be used to measure ambient conditions reliably. However, further work must be done with regards to ethical clearances, sensor limitations and survey sample sizes to implement a functional wristband device using this prototype and validate the correlation between localised skin temperature and subjective comfort levels.

Chapter 2

Introduction {CC}

2.1 Context: Thermal Comfort in University Environments

Thermal comfort is a fundamental requirement in higher education, as it directly impacts cognitive performance, student well-being, and productivity. In environments such as libraries and lecture halls, attendees often face non-uniform thermal conditions. The Health and Safety Executive (HSE) notes that thermal discomfort can lead to “unsafe behaviours”, such as a loss of concentration or fatigue, which are critical during tasks like exams [1].

Current facility management often relies on stationary thermostats; however, these cannot account for the physiological diversity of a student body, where individual metabolic rates and clothing insulation vary significantly. Furthermore, static building-level sensors often fail to capture the micro-climates created by high-occupancy density or localised heat sources common in university laboratories [2].

2.2 Literature Review

2.2.1 Current Standards: The PMV Model

The traditional method for assessing comfort is the Predicted Mean Vote (PMV) model. This numerical index uses a heat-balance equation to predict the average thermal sensation of a large group of people on a standardised scale (Table 2.1).

This methodology is formalised in ISO 7730 [3], which serves as the international benchmark for indoor climate design. While it provides a baseline, research highlights significant limitations in this approach, primarily its reliance on population averages that ignore individual metabolic rates and personal physiological nuances [4].

Table 2.1: ASHRAE Thermal Sensation Scale used in PMV calculations [5]

Value	Sensation
+3	Hot
+2	Warm
+1	Slightly Warm
0	Neutral
-1	Slightly Cold
-2	Cool
-3	Cold

2.2.2 Physiological Indicators and Local Sensation

Thermal comfort is inherently subjective. While internal body temperature typically ranges between 37.0 and 37.8 °C, mean skin temperature seems to act as a more reliable indicator for real-time assessment. Liu et al. [6] demonstrate that by accounting for individual variations such as gender, mean skin temperature becomes a highly dependable metric.

Research into “local” versus “global” sensation suggests that specific body parts exert a disproportionate influence on an individual’s perception of the environment [7]. For instance, facial skin temperature provides high-accuracy data, but monitoring this area is intrusive and impractical for daily academic use. This necessitates a transition to more practical, yet still responsive, extremities such as the limbs.

2.2.3 Transition to Wearables

Recent studies have identified the wrist as having high accuracy for predicting reported thermal sensation among non-intrusive sites [8]. By leveraging the vascularity of the wrist, a wearable system can capture significant physiological data without the invasiveness associated with facial sensors. Specifically, the proximity of the radial artery allows surface sensors to track thermoregulatory changes with high precision [9], justifying the selection of a wristband form-factor.

2.2.4 Weaknesses in the Current State of the Art

Despite the move towards individualised sensing, two major gaps remain:

1. **The Productivity Gap:** Modern research suggests that thermal neutrality (feeling neither hot nor cold) does not automatically equate to peak productivity. Traditional metrics fail to account for the “state of mind” of the occupant [10]. Additionally, environmental factors like CO₂ concentration, which are often overlooked in thermal-only models, have been shown to significantly impair cognitive task performance in indoor workers [11].
2. **The Discursive Gap:** Recent methodologies, such as MADIT (a textual analysis tool used to evaluate how people describe their environment), show that comfort is “discursive”. This means it is a result of the interaction between the person and their specific setting, which static sensors cannot track.

Consequently, there is a clear need for an accessible, real-time monitoring tool tailored to the unique demands of academic environments.

2.3 Project Aim and Objectives

The primary goal of this project is to design and manufacture a wireless wearable device capable of synchronising physiological and environmental data to assess individual thermal comfort in real time. This project addresses the weaknesses identified in current models by correlating local skin and ambient temperature with subjective thermal sensation.

To achieve this, the project involves the following objectives:

- Design and manufacture a compact, wireless sensor node (the “System 2.0”).
- Integrate sensors for skin surface temperature, ambient temperature, relative humidity, and CO₂ concentration.
- Validate the device’s accuracy through experimental trials in UCL facilities, such as the Student Centre and labs.
- Propose a student-centric approach to facility management by comparing collected data against established literature thresholds (32.6 to 33.7 °C).

Chapter 3

System Design {AA, BB}

3.1 Critique of System 1.0

An existing system prototype was provided (Figure 3.1). Table 3.1 summarises the advantages and disadvantages of System 1.0.

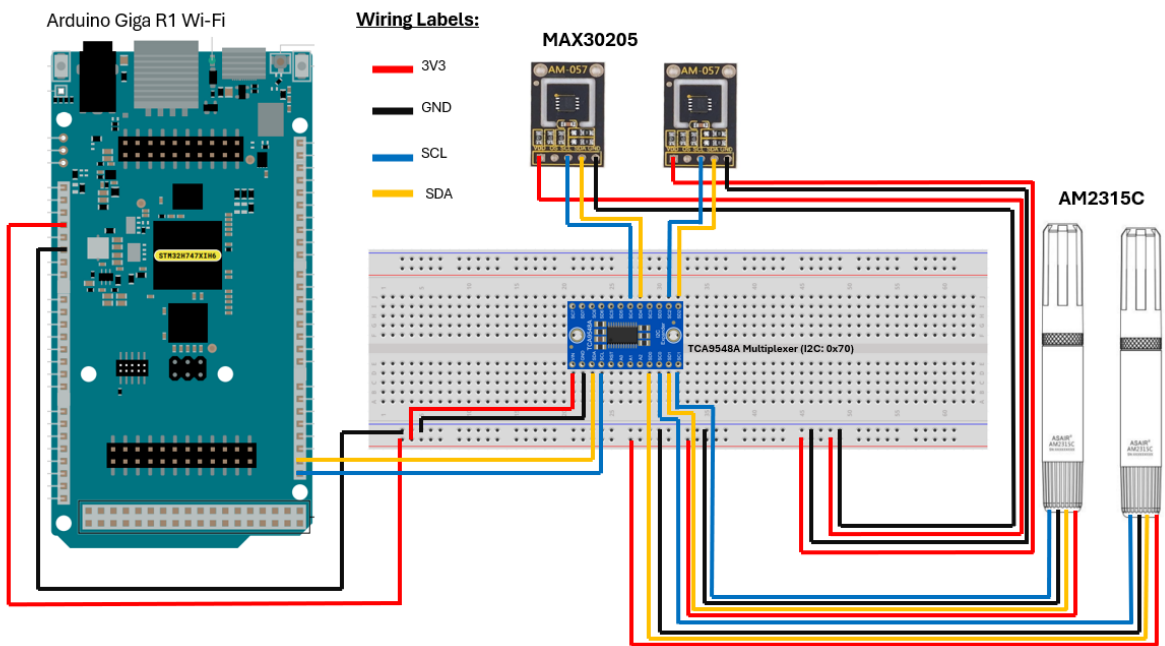


Figure 3.1: Wiring Diagram of System 1.0

The MAX30205 sensors [12] and TCA9548A multiplexer were maintained in the design of System 2.0. A limitation of System 1.0 was the wired design, motivating the necessity of a wireless design.

Table 3.1: Advantages and disadvantages of System 1.0

System 1.0	
Advantages	MAX30205 sensors are highly accurate at skin temperature measurements [12].
	Usage of a multiplexer organises sensors into channels and prevents I2C addresses from clashing.
Disadvantages	Wired design limits the space in which variables can be measured; temperature can vary within a space.
	Wires can break and would need to be repaired.
	AM2315C has an accuracy of $\pm 2\%$ for humidity [13] which can be improved.

3.2 Design Brief for System 2.0

System 2.0 will integrate both a wired and wireless method of transmitting data using microcontrollers.

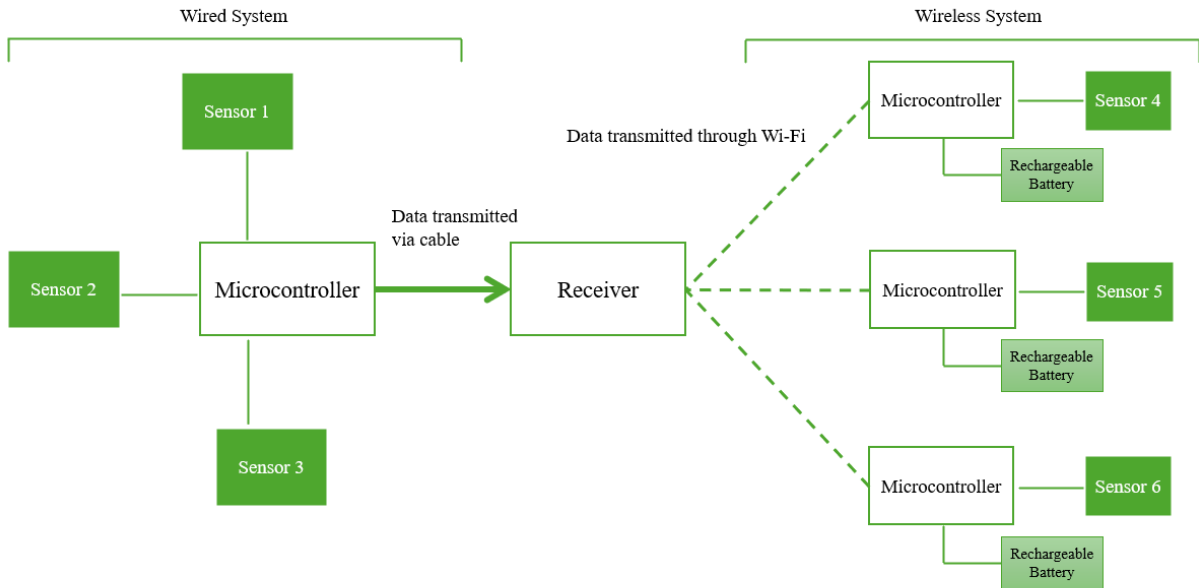


Figure 3.2: Conceptual design in which a wired and wireless system transmit data to a common receiver via cable and Wi-Fi respectively.

- As the individual is expected to be mobile, the skin temperature sensor must be wireless. This can then be made into a wristband.
- To ensure long-term usage, rechargeable batteries must be attached to the wireless microcontrollers.

- Sensors for the wireless system will include:
 - Humidity
 - Skin, surface and ambient temperature
- Sensors for the wired system will include:
 - Humidity
 - Surface and ambient temperature
 - CO₂ concentration

3.3 Final Design and Implementation

All the sensors used and parameters measured within System 2.0 are found in Table 3.2.

Table 3.2: Specifications of SHT45[14], SCD41 [15], MCP9808 [16] and MAX30205 [12].

Parameter Measured	Condition	Accuracy
SHT45		
Ambient Temperature	0–75 °C	±0.1 °C
Humidity	25%–75%	±1.0%
SCD41		
CO ₂ Concentration	400 ppm–5000 ppm	±(40 ppm + 5% of reading)
Ambient Temperature	15 °C–35 °C	±0.8 °C
Ambient Temperature	-10 °C–60 °C	±1.5 °C
Humidity	15 °C–35 °C, 20%RH–65%RH	±6% RH
Humidity	-10 °C–60 °C, 0%RH–100%RH	±9% RH
MCP9808		
Surface Temperature	-40 °C–125 °C	±0.25 °C
MAX30205		
Skin Temperature	15 °C–35.8 °C	±0.3 °C
Skin Temperature	35.8 °C–37 °C	±0.2 °C
Skin Temperature	37 °C–39 °C	±0.1 °C

The wireless system uses the XIAO ESP32C3 [17] to transmit data from the sensors (Figure 3.3) via Wi-Fi to the Arduino Cloud. Wi-Fi is accessible by attaching 2.4GHz antennas to the ESP32C3.

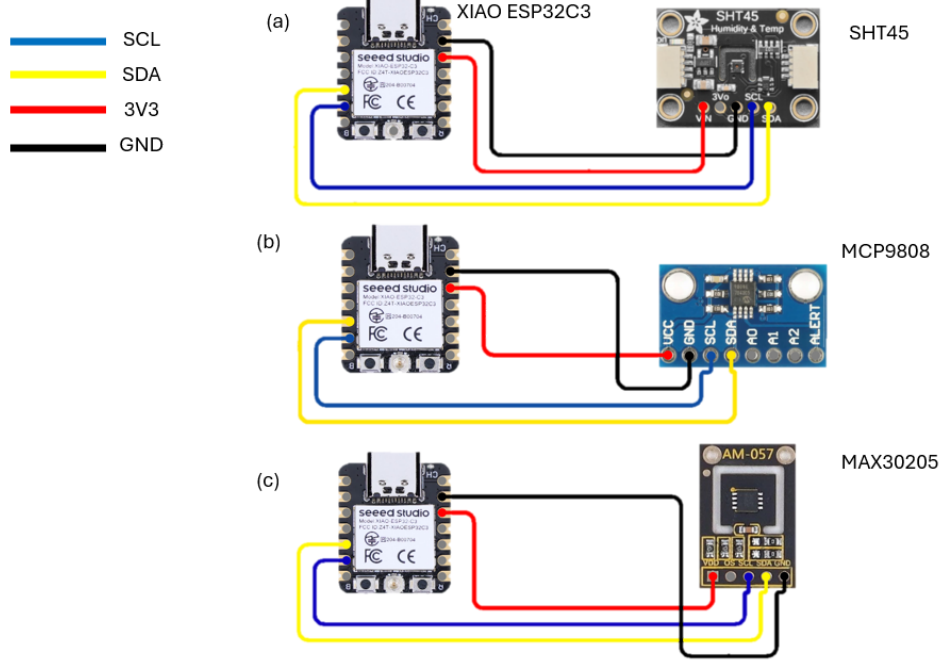


Figure 3.3: Wiring Diagram of the Wireless System. Pins are soldered onto each micro-controller and sensor to allow connection via female-to-female jumper cables.

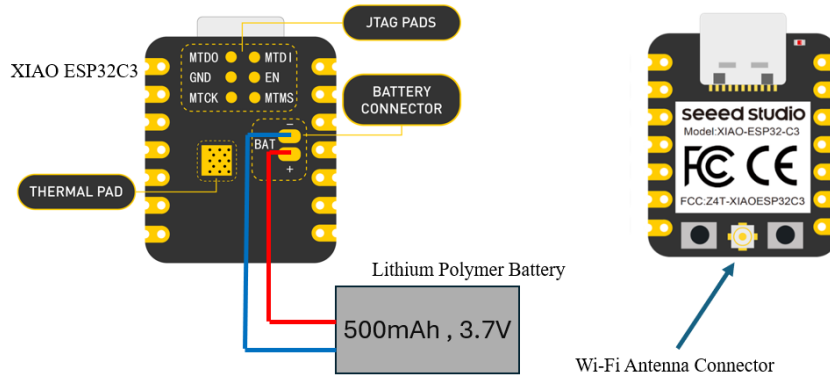


Figure 3.4: Diagram of Battery and Wi-Fi connection to the ESP32C3 [17]. The red wire of the LiPo battery must be soldered to the positive end and the blue (or black) wire is soldered to the negative end.

Data from the sensors is stored as variables within the Cloud and can be extracted as CSV files. This project uses the free version of the Arduino Cloud, meaning only 2 microcontrollers are registered per account.

An account for the Arduino Cloud can be created using email and password. It will prompt the installation of the Arduino Cloud Agent, which is required for connection. To register the microcontrollers, the ESP32C3 must be connected to a computer via a USB-C to USB cable. On the Arduino Cloud website, “Add Device” is selected. A security key

will be given that must be downloaded and kept secure.

The MAC address of each ESP32C3 will have its own password provided by an I.T. Systems Administrator to connect to the UCL IoT Wi-Fi.

A ‘Thing’ is created and has the device associated with it. This means that data transmitted from the microcontroller will be sent to that Thing. Variables are created within the Thing to organise the data.

A dashboard can be created to monitor the variables from each sensor. The data monitored throughout the day can be stored as a CSV file.

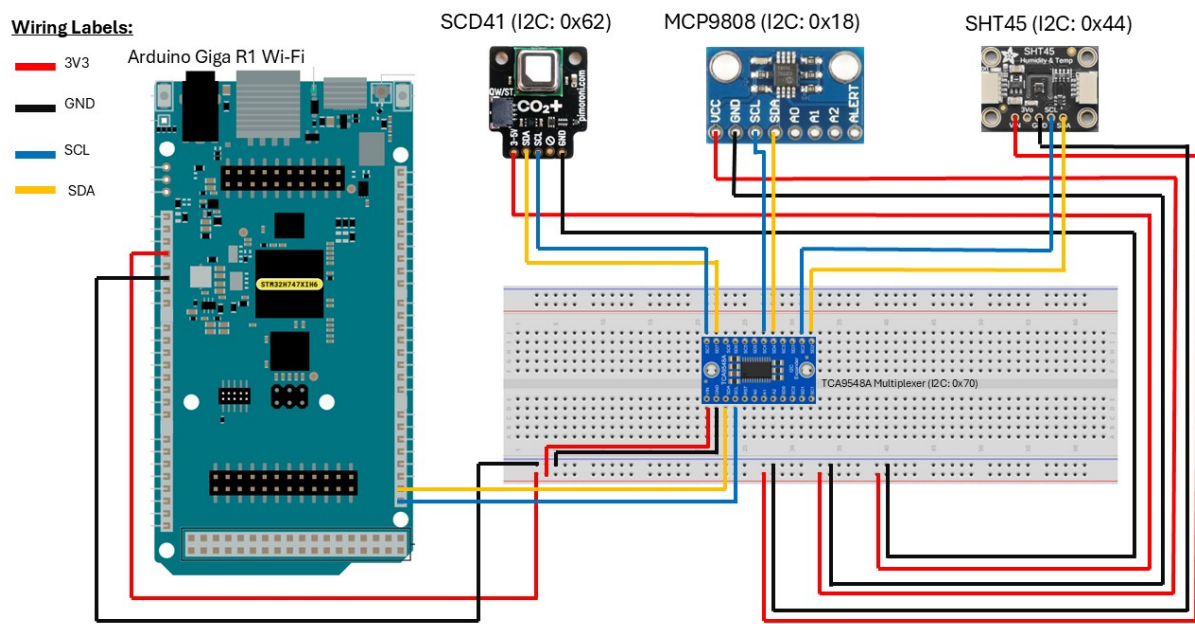


Figure 3.5: Wiring Diagram of the Wired System.

The wired system (Figure 3.5) uses the Arduino Giga R1 Wi-Fi connected to sensors and transmits data via Serial to the Arduino IDE. The MAX30205 is replaced by the SCD41 to measure CO₂ concentration. A multiplexer allows for the organisation of sensors into specific channels to prevent I²C addresses clashing.

3.4 Data Recording Interface

Sensor data was acquired and processed via a Python program through the PyQt5 module. The program established a serial connection to the Arduino Giga R1 Wifi and received six sensor values per transmission. Data was logged at a sampling rate of 0.2 Hz.

Live visualisation, achieved via matplotlib, was implemented to monitor the sensors. Temperature data from the MCP9808, SHT45, and SCD41 were overlaid on a single temperature plot. Humidity data from the SHT45 and SCD41 were similarly combined with CO₂ readings being separated. Each variable was separated into individual tabs.

Following data collection, the live graphs could be stopped and all data was stored into a JSON file with corresponding timestamps for analysis.

Chapter 4

Experimental Design {AA, BB, CC}

4.1 Methodology

4.1.1 Validation of System Performance

To ensure System 2.0 reliability, a comparative analysis was conducted between the heat-generating device [18], the wireless wearable node and the wired system.

1. In the MPBE laboratory, both systems were set up so that the temperature sensors made direct contact with the heat source.
2. The heat source temperature was increased in intervals of 2 °C from 27–43 °C.
3. After each change in source temperature, a 5-minute adjustment period was allowed to account for initial output fluctuations observed in early trials.

4.1.2 Investigation into environmental variables and their impact on thermal comfort

Environmental data (temperature, humidity and CO₂ concentration) were collected at 4 locations: Student Centre L2 and L4, the Medical Physics and Biomedical Engineering Lab, and the Christopher Ingold XLG1 Lecture Theatre. The experimental procedure was conducted as follows:

1. Both the wired and wireless systems were set up at each fixed location.
2. Using the aforementioned program, environmental data was recorded over a 30-minute interval following the same 5-minute adjustment period.
3. Subjective data was gathered via Google forms for all locations immediately after data collection for the lecture theatre and laboratory. However, due to participant availability and time constraints, subjective data for the Student Centre was gathered the following day at midday.

4.2 Results and Analysis

4.2.1 System Performance and Sensor Validation

Table 4.1: Error Analysis of Wired and Wireless Temperature Sensors

Sensor Type	Mean Error (°C)	Max Error (°C)	RMSE (°C)	Correlation Coefficient (R)
Wireless	1.84	4.31	2.45	0.993
Wired	2.18	4.25	2.64	0.989

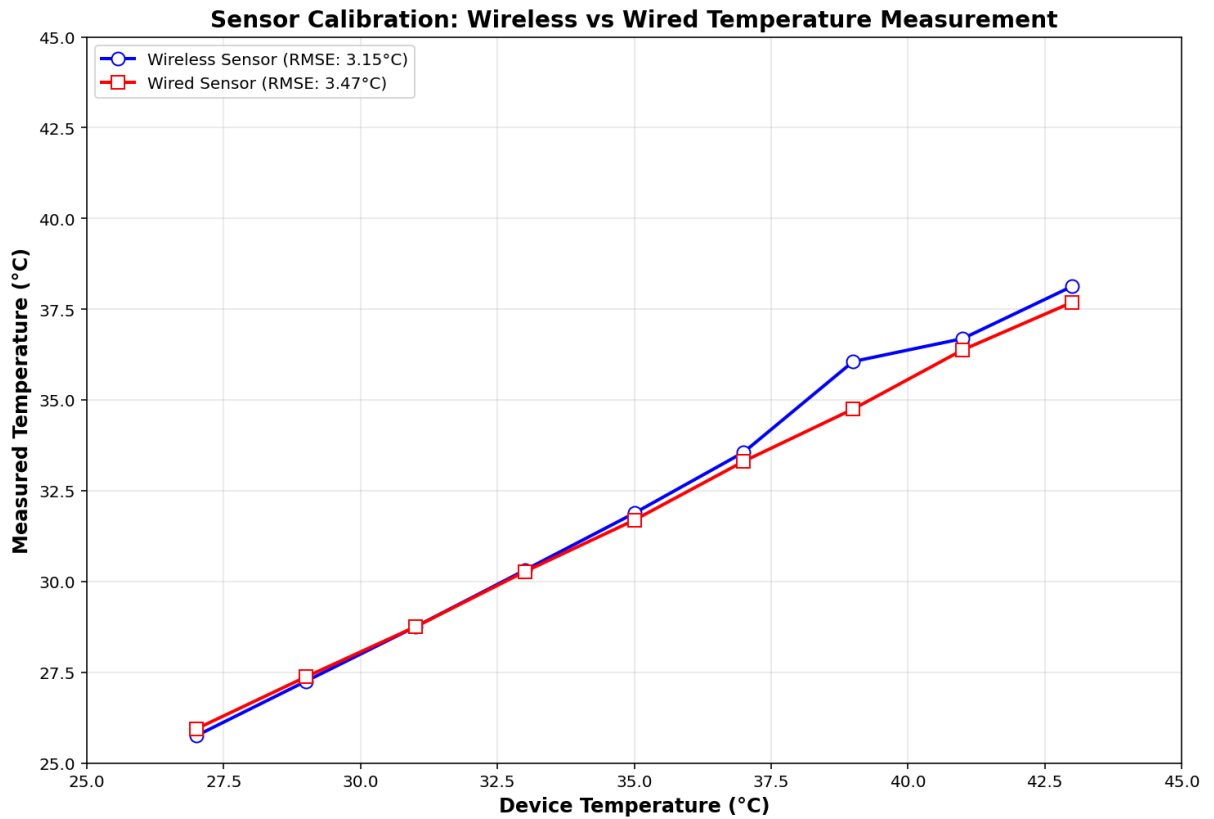


Figure 4.1: A graph comparing the wired and wireless prototypes against a reference heat source.

Analysis: The results indicate both systems followed the increase in heat source temperature with high correlation coefficients and similar mean errors, proposing the wireless prototype as a suitable alternative for environmental monitoring. However, Figure 4.1 shows error increased at higher temperatures for both systems, emphasising the influence of environmental conditions on sensor calibration.

4.2.2 Environmental Data Summary

Due to time constraints, for all locations, only one dataset was obtained; therefore, the standard deviation represents the variation within a single data collection period.

Table 4.2: Temperature Readings Across All Locations from Both Systems (Mean $\pm$ Standard Deviation)

Location	System	SHT45 ($^{\circ}\text{C}$)	MCP9808 ($^{\circ}\text{C}$)	SCD4x ($^{\circ}\text{C}$)	Average ($^{\circ}\text{C}$)
Student Centre (L2)	Wired	24.8 (0.14)	24.81 (0.34)	24.48 (0.58)	24.70 (0.15)
	Wireless	22.89 (0.15)	23.64 (0.40)	–	23.27 (0.38)
Student Centre (L4)	Wired	26.68 (1.06)	25.89 (0.62)	23.71 (0.17)	25.43 (1.26)
	Wireless	23.20 (0.32)	22.78 (0.04)	–	22.99 (0.21)
Laboratory	Wired	22.64 (0.00)	23.11 (0.13)	22.60 (0.07)	22.78 (0.23)
	Wireless	22.17 (0.19)	22.99 (0.15)	–	22.58 (0.41)
Lecture Theatre	Wired	20.86 (0.64)	21.05 (0.45)	20.47 (0.53)	20.79 (0.24)
	Wireless	21.43 (0.59)	19.70 (0.29)	–	20.57 (0.87)

Note: The average column represents the mean of all available sensors for each row. Dashes (–) indicate the absence of a sensor from a system.

Table 4.3: Humidity variation by both the wired and wireless prototypes in all 4 locations

Location	System	SHT45 (%)	SCD4x (%)
Student Centre (L2)	Wired	44.2 (0.9)	45.4 (1.9)
	Wireless	46.7 (0.4)	–
Student Centre (L4)	Wired	41.9 (1.7)	44.4 (0.7)
	Wireless	45.6 (1.1)	–
Laboratory	Wired	44.8 (0.0)	43.2 (0.1)
	Wireless	46.4 (0.5)	–
Lecture Theatre	Wired	43.2 (0.6)	42.0 (1.9)
	Wireless	41.8 (0.9)	–

Table 4.4: Summary of the CO₂ measurements by the wired system in all 4 locations

Location	Mean (ppm)	σ (ppm)	Minimum (ppm)	Maximum (ppm)
Laboratory	446.1	57.3	408	532
Lecture	585.7	105.1	402	786
Student Centre (L2)	802.9	65.8	696	1047
Student Centre (L4)	771.5	12.3	745	793

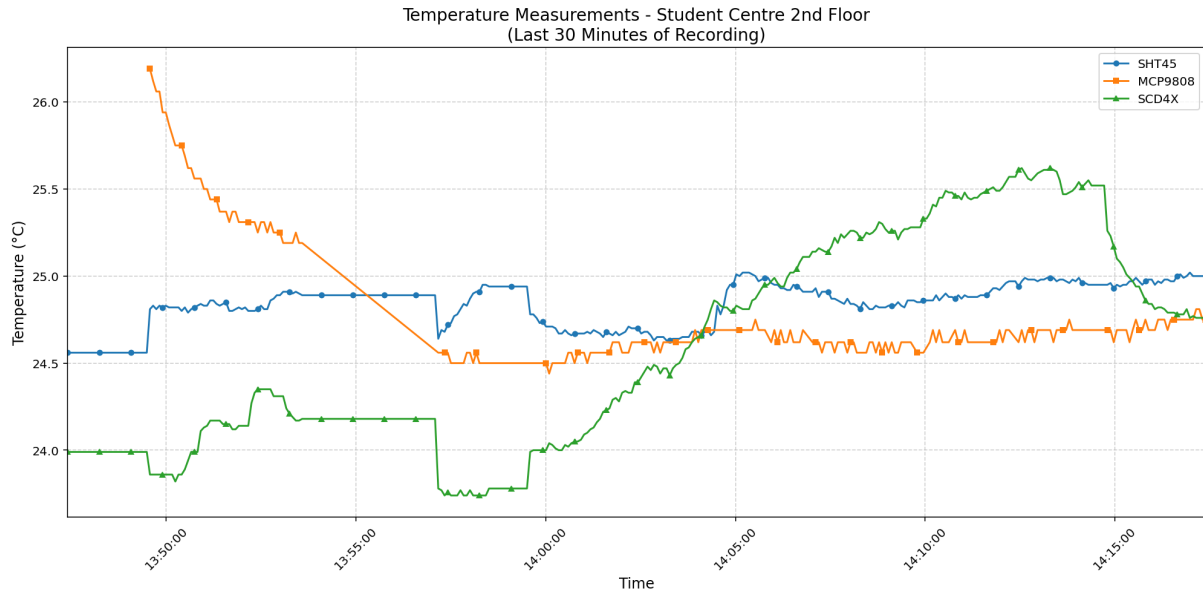


Figure 4.2: Live visualisation of the temperature sensors for 30-minute data collection period. Graph highlights the reliability issues of the MCP9808 sensor.

Temperature Analysis

The wired system recorded its highest average temperature (25.43 °C) on the Student Centre fourth floor (L4), exceeding the second floor (L2) by 0.73 °C due to higher altitude and greater sun exposure. L4 also showed greater temperature variability ($\sigma = 1.26$) than L2 ($\sigma = 0.15$), suggesting less stable conditions. The laboratory was the most stable environment due to low occupancy.

The lecture theatre recorded the lowest temperature (20.57 °C via wireless) and was the only location where a wireless sensor (21.43 °C by SHT45) recorded a higher temperature than its wired counterpart. This is likely because the wireless system was placed at the front and the wired system at the back. Across all locations, the wireless system recorded lower average temperatures than the wired system, indicating different calibration times. Intermittent data loss (Figure 4.2) reduced the reliability of the MCP9808 sensor, suggesting its calculated mean and standard deviation may not fully represent actual conditions.

Humidity Analysis

Outside of the laboratory, which showed consistent humidity levels likely due to its good ventilation and low occupancy, more occupied spaces like the Student Centre L2 and lecture theatre exhibited humidity variation as high as $\sigma = 1.9\%$.

CO₂ Analysis

As expected, the highest CO₂ levels were recorded in the Student Centre L2 (mean = 802.9 ppm & peak of 1047 ppm) due to its high foot traffic, limited ventilation from the lack of open windows and the relatively high occupancy throughout the day. The wide variation at this location also represents the fluctuating congestion levels indicative of a busy area. L4 showed lower levels of CO₂, indicating better ventilation.

In contrast, the laboratory recorded the lowest CO₂ levels (446 ppm), consistent with the low occupancy observed. The lecture theatre showed moderate CO₂ levels (mean = 771.5 ppm) due to its space and early morning measurements.

4.2.3 Subjective Survey Data

Subjective data was collected at the UCL Student Centre ($n = 7$), XLG1 Lecture Theatre ($n = 22$), and MPBE laboratory ($n = 3$). Laboratory responses were excluded from analysis due to insufficient sample size. The following analysis focuses on the data collected at the Student Centre and lecture theatre to establish the “Discursive Gap” between perceived comfort and physical sensor readings.

Student Centre Findings

- **Thermal Sensation:** 71.4% of participants reported a sensation of +1 (slightly warm) on the PMV scale.
- **Comfort Levels:** 57.1% reported feeling “Comfortable,” while 28.6% remained “Neutral.”
- **Temperature Estimation:** Predicted values showed high variance, ranging from 20 °C to 37 °C.

Table 4.5: Frequency of Subjective Temperature Estimations (Student Centre (L2+L4))

Temperature Value (°C)	Frequency
20–24	3
25–29	2
30–34	1
35–39	1

Lecture Theatre Findings

- **Thermal Sensation:** Responses were widely distributed: 47.6% neutral (0), 28.6% slightly warm (+1), and 23.8% slightly cool (−1).
- **Comfort Levels:** Increased discomfort compared to the Student Centre, with 23.8% reporting they felt “Uncomfortable.”
- **Perceived vs. Actual:** As shown in Table 4.6, the most frequent estimate was 20 °C ($n = 4$), despite the broader spread of actual ambient readings.

Table 4.6: Frequency of Subjective Temperature Estimations (Lecture Theatre)

Temperature Value (°C)	Frequency
15–19	6
20	5
21–23	6
24–25	4
38	1

4.3 Data Analysis

4.3.1 Comparison Against Literature Thresholds

Literature (Section 2.2.2) defines a physiological mean skin temperature comfort threshold between 32.6–33.7 °C [6]. While ethical constraints prevented live skin monitoring via the MAX30205 sensor, recorded ambient conditions across all surveyed locations were evaluated against the ISO 7730 and PMV models introduced in Section 2.2.1.

4.3.2 Comparative Analysis of Results

The data highlights a thermal gradient correlating with the shifting subjective sensations of participants:

- **Validation of PMV Thresholds:** In the Student Centre (L4), temperatures averaged 25.43 °C, coinciding with a 71.4% reported sensation of +1 (Slightly Warm). Conversely, the Lecture Theatre averaged 20.57 °C, where 23.8% of participants reported −1 (Slightly Cool). These results validate the ISO 7730 [3] model, which predicts discomfort when temperatures exceed 25 °C or fall below 21 °C in standard indoor conditions.
- **Quantifying the Discursive Gap:** Tables 4.5 and 4.6 provide empirical evidence for the “Discursive Gap” (Section 2.2.4). Despite stable ambient means, 28.6% of Student Centre participants estimated the temperature to be above 30 °C (with one reaching 38 °C), while Lecture Theatre estimations dropped to 15 °C. This 22 °C perceived variance confirms that subjective perception is a poor environmental monitoring tool, reinforcing the necessity for System 2.0’s precision sensors.
- **Personal Preference vs. Static Standards:** The Lecture Theatre showed the widest distribution of sensations (ranging from −1 to +1) at a constant 20.57 °C. This spread suggests that personal physiological thresholds vary significantly more than the static population averages used in the PMV model. This directly validates the project’s core aim: building-wide thermostats cannot satisfy a diverse student body, necessitating the individualised wearable approach of System 2.0.

Chapter 5

Conclusions {AA, BB, CC}

Our investigation successfully developed a wireless wearable system (System 2.0) capable of synchronising physiological and environmental data. By integrating high-precision humidity and CO₂ sensing, the prototype addressed the “Productivity Gap” and demonstrated high potential for mobile data collection, with a mean thermal error (1.84 °C) comparable to the wired reference.

Environmental analysis across four UCL facilities revealed significant micro-climate variations. The Student Centre exhibited the highest CO₂ levels (802.9 ppm) and a 0.73 °C vertical temperature gradient between floors L2 and L4, likely driven by solar gain and ventilation. While altitude was not isolated as a primary variable, this observed gradient suggests it contributes to thermal sensation.

Subjective survey data ($n = 32$) highlighted a clear “Discursive Gap”; while ambient conditions (25.43 °C) validated the ISO 7730 PMV model, participant temperature estimations varied wildly (15 °C to 38 °C). This reinforces that human perception is an unreliable metric for facility management and confirms that comfort is a multi-variant outcome influenced by the interplay of temperature and CO₂.

Due to ethics approval delays and time constraints, the MAX30205 was not utilised for live human skin monitoring, and software limits restricted concurrent data streams. Future iterations (System 3.0) should focus on:

- Securing ethical clearance to correlate localised skin temperature with subjective sensation.
- Expanding the demographic sample size for higher statistical significance.
- Investigating alternative medical-grade thermistors or non-contact IR sensors to resolve MCP9808 reliability issues.
- Integrating a barometric pressure sensor (e.g., BMP280 [19]) to mathematically correlate altitude with the observed vertical temperature gradients across multi-level facilities.

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Appendix A: Appendices

A.1 Meeting Minutes

Project Management Note: The group maintained a consistent schedule, meeting at least once per week from 14/01/2026 to 23/03/2026. While all members (AA, BB, CC, DD, EE) were present for the initial project definition and system architecture phases, subsequent experimental sessions and lab-based hardware debugging were primarily conducted by AA and BB. This was due to significant timetable clashes and a strategic division of labour, where CC, DD, and EE focused on report writing and assisting experimental analysis and design as made apparent throughout the chapter headings. All members contributed to the final data analysis and report review.

Phase 1: Project Initiation & Procurement (Weeks 1–3)

Board Member Presence: Present **Average Attendance:** 100% (AA, BB, CC, DD, EE)

Agenda & Discussion:

- **Analysis of System 1.0:** Documented the failures of System 1.0 and brainstormed improvements for the new prototype. Discussed the move to medical-grade sensors (MAX30205) and high-precision CO₂ monitoring (SCD41).
- **Research and Experimental Design Planning:** Researched sources relevant to thermal comfort studies and discussed variables required for the experimental design to meet the project's primary aims.

Actions arising: AA to initialise procurement via the department; CC to establish the BibTeX referencing library.

Board Member Feedback (Pilar): Offered guidance on finances and gave feedback on the literature review which was implemented by CC

Phase 2: System Design & Firmware Development (Weeks 4–6)

Board Member Presence: Week 4 only **Average Attendance:** 100% (AA, BB, CC, DD, EE)

Agenda & Discussion:

- **Prototype Discussion:** Evaluated the use of the TCA9548A Multiplexer to resolve address conflicts and assessed ESP32C3 WiFi stability for cloud-based data streaming.
- **Financial Review:** Researched components and drafted a financial review within the £250 budget; commenced ordering for the final prototype.
- **Lab Sessions:** The full group attended soldering training in the lab and collectively tested System 1.0 to gain a technical understanding of the existing code and hardware requirements for the new build.

Actions arising: BB to finalise ESP32 firmware; EE to draft the electrical circuit schematics.

Board Member Feedback (Pilar): Assistance in creating the project timeline and splitting up tasks amongst the group.

Phase 3: Experimental Data Collection (Weeks 7–8)

Board Member Presence: Not Present **Average Attendance:** 60–80% (AA, BB, CC, DD)

Agenda & Discussion:

- **Data Protocol:** Standardised the 30-minute collection window and 5-minute sensor adjustment period.
- **Subjective Survey:** Finalised the Google Form (BB) using the ISO 7730 PMV sensation scale based on research conducted by CC.

Actions arising: Conduct data collection at UCL Student Centre (AA, BB), Laboratory (AA, BB, CC), and Lecture Theatre (AA, BB, CC, DD).

Supervisor Check-in: Group worked independently with minimal intervention from Pilar during this phase.

Phase 4: Final Synthesis & Poster Design (Weeks 9–10)

Board Member Presence: Not Present **Average Attendance:** 60% (AA, BB, CC)

Agenda & Discussion:

- **Results Interpretation:** Interpreted the 22 °C spread in perceived temperature vs. actual sensor data (The Discursive Gap).
- **Conclusions:** Jointly drafted the final conclusions to reflect on productivity and thermal sensation outcomes.

Actions arising: Each member assigned a chapter for final peer-review; AA to contact supervisor for poster printing assistance.

Supervisor Check-in: Pilar provided final guidance on the poster layout and confirmed assistance with printing logistics.

A.2 Financial Summary

The following table details the hardware procured for the development of System 1.0 and System 2.0, including the specific purpose of each component within the project architecture.

Component	Supplier	Qty	Total (£)	Purpose
Arduino Giga R1 WiFi	The Pi Hut	1	65.00	Primary wired hub, selected for high memory and multiple I2C ports to handle large data streams.
MAX30205 Temp Sensor	Amazon	2	47.08	Medical-grade digital sensor (± 0.1 °C) used for precise human skin temperature monitoring.
SCD41 CO ₂ Sensor	The Pi Hut	1	43.50	CO ₂ concentration sensor, chosen over estimated CO ₂ models to guarantee data accuracy.
Adafruit SHT45	The Pi Hut	2	26.00	High-precision ambient temperature and relative humidity sensor.
500mAh 3.7V LiPo Battery	The Pi Hut	3	18.00	Portable power supply for wireless sensor nodes (around 6 hours of operation).
XIAO ESP32C3	CPC Farnell	3	12.54	Microcontrollers for wireless data transmission to Cloud.
MCP9808 Temp Sensor	The Pi Hut	3	14.40	High-accuracy digital sensor (± 0.25 °C) for general ambient and surface temperature monitoring.
TCA9548A Multiplexer	The Pi Hut	1	6.70	Resolves I2C address conflicts, enabling simultaneous data collection from multiple sensors.
Shipping Costs	Various	-	13.89	Delivery fees across multiple hardware suppliers.
Poster	Prontaprint	1	Paid By Supervisor	For Poster Session.
Total Project Cost			247.11	
Cost per Member ($n = 5$)			49.42	

Table 5.1: Hardware Procurement and Financial Breakdown

A.3 Risk Assessments

Hazard	Risk	Level	Control Measures
Burning (Heat)	Contact with conducting parts	Low	Calibrate properly; avoid contact with active parts.
Electrification	Electric shock from exposed parts	Low	Equipment inspection; cover parts with insulators.
Room Temp.	Heat-related illness	Low	Team monitoring; light clothing; water kept outside room.
Soldering Iron	Burns from tip or molten solder	Med	Use safety stand; wear gloves; obtain proper training.
Solder Fumes	Respiratory hazards from flux	Med	Use fume extractors; consider lead-free solder.

A.4 Technical Documentation

The integration of the **TCA9548A I2C Multiplexer** was a critical technical decision for the data collection phase; the component allowed the system to bypass the physical limitations of the I2C protocol when handling high-precision instrumentation. Key advantages included:

- **Address Conflict Mitigation:** The multiplexer enabled the simultaneous use of the **SHT45**, **MAX30205**, and **MCP9808** sensors. Since several of these devices share fixed or limited I2C address options, the TCA9548A provided the necessary bus segmentation to prevent data collisions.
- **Data Synchronisation:** By centralising the sensor suite on a single controlled interface, the wired reference data and wireless wearable data were sampled at an identical frequency. This temporal alignment was critical for the validity of the correlation analysis presented in the results.

A.5 Project Repository and Source Code

All source code developed for this project, including the **C++** firmware for the XIAO ESP32C3 and Arduino Giga R1, as well as the Python scripts used for data processing and live visualisation, is hosted on GitHub.

GitHub Repository:

<https://github.com/mhanyasoos/group-15-project>